

Project Title	SentinelNet-AI-Powered Network Intrusion Detection System (NIDS)
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SentinelNet-AI-Powered Network Intrusion Detection System (NIDS)

ABSTRACT

SentinelNet – AI-Powered Network Intrusion Detection System (NIDS) is designed to provide a comprehensive and intelligent solution for monitoring and securing computer networks against cyber threats. With the increasing sophistication and frequency of cyber-attacks, traditional rule-based intrusion detection systems often fail to detect novel or evolving threats. To address this limitation, SentinelNet leverages **advanced machine learning techniques** to classify network traffic in real time as either normal or malicious. The system utilizes benchmark datasets such as NSL-KDD and CICIDS2017, which offer a wide variety of attack scenarios, traffic features, and labeled samples for effective model training. Through preprocessing, feature selection, and dimensionality reduction, the system identifies the most relevant network attributes that contribute to accurate intrusion detection. SentinelNet employs multiple machine learning models, including ensemble learning methods, to enhance detection accuracy, reduce false positives, and ensure robustness against diverse types of attacks such as Denial-of-Service (DoS), probing, and ransomware. By providing **real-time alerts** and actionable insights, the system enables proactive threat mitigation, helping organizations safeguard sensitive data and maintain network integrity. Overall, SentinelNet represents a scalable, efficient, and AI-driven approach to modern network security challenges, bridging the gap between traditional NIDS approaches and the demands of increasingly complex cyber environments.

Keywords: SentinelNet, Cyber Threats, Machine Learning, Network Traffic, Preprocessing, Ensemble Learning, Real-time Alerts

INTRODUCTION

In an increasingly interconnected digital world, the need for robust network security has become paramount. With the proliferation of devices, services, and data, modern networks face a constant barrage of sophisticated and evolving cyber threats. Traditional security measures, such as firewalls and signature-based intrusion detection systems (IDS), are often reactive and struggle to keep pace with new, "zero-day" attacks that lack pre-defined signatures. This creates a critical vulnerability for organizations and individuals alike.

To address these limitations, this project introduces SentinelNet, an innovative AI-powered Network Intrusion Detection System (NIDS). SentinelNet leverages the power of artificial intelligence (AI) and machine learning (ML) to move beyond traditional, static defense mechanisms. Instead of simply looking for known attack patterns, our system learns and adapts by analyzing vast amounts of network traffic data to establish a baseline of "normal" behavior. Any deviation from this baseline is flagged as an anomaly, enabling the detection of both known and unknown threats with a high degree of accuracy and a reduced number of false positives.

The core objective of SentinelNet is to provide a proactive, intelligent, and scalable solution to enhance network security. By employing advanced algorithms, the system can autonomously identify and respond to malicious activities, such as Denial of Service (DoS) attacks, probes, and malware, in real time. This documentation outlines the methodology, implementation, and evaluation of the SentinelNet NIDS, demonstrating its potential to revolutionize the landscape of network defense and provide a more resilient and adaptive security posture against the ever-growing threat landscape.

Following this introduction, the remainder of this documentation is structured as follows: Section 2 reviews **related work** in AI-based NIDS. Section 3 details the **methodology** and **system architecture** of SentinelNet, including data processing and the selected machine learning models. Section 4 presents the **experimental setup, results, and performance evaluation** using benchmark datasets. Finally, Section 5 concludes the document and proposes avenues for **future work**.

1. LITERATURE

The field of Network Intrusion Detection System (NIDS) has evolved in response to the ever-changing landscape of cyber threats. This section provides an overview of the existing approaches and highlights their limitations, providing the context and motivation for the SentinelNet project.

1.1. Traditional Intrusion Detection Systems (IDS)

Historically, NIDS have been categorized into two main types:

- **Signature-Based IDS (SIDS):** These systems rely on a database of pre-defined attack signatures or patterns.
 - **Strengths:** Highly effective at detecting known attacks with minimal false positives.
 - **Limitations:** The primary weakness is their inability to detect **zero-day attacks** or novel threats for which no signature exists, making them reactive.
- **Anomaly-Based IDS (AIDS):** These systems work by first establishing a baseline of "normal" network behavior. Any activity that deviates significantly from this baseline is flagged as a potential intrusion.
 - **Strengths:** Capable of detecting new and unknown attacks, including zero-day exploits, without prior knowledge of their signatures
 - **Limitations:** They are often prone to a high rate of **false positives**.

1.2. Machine Learning in NIDS

The limitations of traditional methods have driven researchers to explore more intelligent, adaptive approaches using Artificial Intelligence (AI) and Machine Learning (ML)

- **Early ML-based NIDS:** Early research applied classic ML algorithms like Support Vector Machines (SVM), Decision Trees, and K-Nearest Neighbors (KNN) to intrusion detection. These methods offered better detection rates but often struggled with high-dimensional datasets and required extensive **manual feature engineering**, a time-consuming process that can impact performance.

1.3. SentinelNet's Position

Building upon the progress and recognizing the gaps in the existing literature, SentinelNet is designed to provide a more effective and practical solution. Our project aims to leverage machine learning models like random forest, logistic regression and xgboost to address the high false positive rates of anomaly-based systems and the limitations of traditional methods in detecting novel threats. We will focus on creating a system that is not only highly accurate but also computationally efficient and robust against modern evasion techniques, pushing the boundaries of what is possible in AI-powered network security.

2. METHODOLOGY

2.1. DATABASE DESCRIPTION

The effectiveness of SentinelNet relies on training with diverse, high-quality benchmark datasets. We utilized two core datasets to ensure robustness against both historical and modern cyber threats: **NSL-KDD** and **CICIDS2017**.

NSL-KDD

Since 1999, the **Knowledge Discovery and Data Mining (KDD'99)**, widely used data set for the estimation of **anomaly detection** techniques. Some investigators examined KDD99 but results were poor execution on the anomaly detection approach so the best solution was using a new dataset which is **NSL-KDD**. This dataset consists of chosen records of the all KDD data set. The Training dataset consists of **(125,973)** and test dataset consists of **(22,544)** samples each of sample contains **41 attributes** either **attacks** or **normal**. Attacks in this dataset are split into: **Root to Local (R2L)**, **User to Root (U2R)**, **Denial of Service (DOS)**, **Probe Attacks**. The following shows some explanations for these attack types: -

1. **Dos**: It is a class of attack where the attacker restricts processing time of the resources so as to avoid the real user from obtaining those resources.
2. **R2L**: Attackers are not allowing access from a remote system. R2L attack is both categorized under network based and host-based (**NIDS**).
3. **U2R**: the attacker tries to gain the password of the user and then get into the system as a legitimate user and retrieve the data.
4. **Probe**: The Attacker will examine the network to collect information and would make some penetrations in the next time.

Schema: Duration, protocol_type, service, flag, src_bytes, dst_bytes, ..., label.

CICIDS2017

Since the inception of CICIDS2017 dataset the dataset started attracting researchers for analysis and developments of new models and algorithms. According to the author of CICIDS2017, the dataset spanned over eight different files containing five days normal and attacks traffic data of Canadian Institute of Cybersecurity. A short description of all those files are presented in Table 1.

It can be seen from Table 1 that the dataset contains attack information as five days traffic data. Thursday working hour afternoon and Friday data are well suited for binary classification. Similarly, Tuesday, Wednesday and Thursday morning data are best for designing multiclass detection model. However, it should be noted that a best detection model should be able to detect attacks of any type. Therefore, to design such a typical IDS, the traffic data of all the day should be merged together to form a single dataset to be used by IDS. This is exactly we followed to merge these files.

Table 1: Description of files containing CICIDS2017 dataset

Name of Files	Day Activity	Attacks Found
Monday-WorkingHours.pcap_ISCX.csv	Monday	Benign (Normal human activities)
Tuesday-WorkingHours.pcap_ISCX.csv	Tuesday	Benign, FTP-Patator, SSH-Patator
Wednesday-workingHours.pcap_ISCX.csv	Wednesday	Benign, DoS GoldenEye, DoS Hulk, DoS Slowhttptest, DoS slowloris, Heartbleed
Thursday-WorkingHours-Morning-WebAttacks.pcap_ISCX.csv	Thursday	Benign, Web Attack – Brute Force, Web Attack – Sql Injection, Web Attack – XSS
Thursday-WorkingHours-Afternoon-Infiltration.pcap_ISCX.csv	Thursday	Benign, Infiltration
Friday-WorkingHours-Morning.pcap_ISCX.csv	Friday	Benign, Bot
Friday-WorkingHours-Afternoon-PortScan.pcap_ISCX.csv	Friday	Benign, PortScan
Friday-WorkingHours-Afternoon-DDos.pcap_ISCX.csv	Friday	Benign, DDoS

2.2. PREPROCESSING

NSL-KDD Dataset Preprocessing:

- 1. Data Collection:** The preprocessing pipeline began by loading the raw data from the standard files: **KDDTrain+_20Percent.txt** and **KDDTest-21.txt**.
- 2. Data Cleaning:** Data cleaning involved removing the redundant difficulty column from both the training and testing sets. Furthermore, the dataset was checked for and cleaned of any missing values by dropping rows containing NaN values (.dropna()), and duplicate records were also checked for and removed (.drop_duplicates()).
- 3. Data Transformation:** The primary transformation involved **Encoding Categorical Data: Label Encoding** was applied to the string-based categorical columns, specifically **protocol_type**, **service**, and **flag**. Additionally, a critical **Feature Engineering** step was performed to convert the multi-class **label** column into a **binary target variable, attack**, where 'normal' was mapped to 0 and all attack classes were mapped to 1.
- 4. Data Reduction:** Dimensionality Reduction via feature removal was performed by explicitly dropping the num_outbound_cmds column from the dataset.
- 5. Imbalance Handling:** Synthetic Minority Over-sampling Technique (SMOTE) was applied to the training data to balance 18 of the 22 classes in multiclass classification and attack feature in binary classification.

6. **Scaling: StandardScaler** (Z-score normalization) was applied to all numerical features.
7. **Data Split:** The data was prepared for model evaluation by separating the final preprocessed feature set (X) from the newly engineered binary target column (y or attack). Data split into Training (80%) and Testing (20%).

CICIDS2017 Dataset Preprocessing:

1. **Data Collection:** The process started by loading the network flow data from a single CSV file, specifically **Friday-WorkingHours-Afternoon-DDos.pcap_ISCX.csv** for Binary classification and **Wednesday-WorkingHours.pcap_ISCX.csv** for Multiclass classification.
2. **Data Cleaning:** Extensive cleaning was performed for robustness. First, all column names had leading/trailing whitespace stripped. Next, all problematic **infinite values** (np.inf and -np.inf) were replaced with **NaN**. Subsequently, all rows containing either the converted or original **NaN** values were **dropped** (df.dropna()). Finally, all **duplicate rows** were also removed from the dataset.
3. **Data Transformation:** The main transformation was the **Encoding Categorical Data** for the target variable: **Label Encoding** was applied to the **label** column to convert the attack names into numerical classes suitable for machine learning algorithms.
4. **Data Reduction: Feature Selection** was performed by dropping several identifying columns that are not useful for training a general intrusion detection model, including **Flow ID**, **Source IP**, **Source Port**, **Destination IP**, **Destination Port**, and **Timestamp**.
5. **Imbalance Handling:** The **Synthetic Minority Over-sampling Technique (SMOTE)** was applied to the training data to balance the minority 'DDoS' and 6 attack classes in multiclass classification.
6. **Scaling: StandardScaler** (Z-score normalization) was applied to all numerical features.
7. **Data Split:** The final preprocessed data was prepared for modeling by separating the cleaned feature set (X) from the encoded target variable (y) before splitting the dataset into training and testing subsets using **train_test_split**. Data split into Training (80%) and Testing (20%).

2.3. NOVELTY

The novelty of SentinelNet lies in its systematic, comparative, and balanced approach:

1. **Hybrid Dataset Validation:** Testing the same ensemble ML approach on both KDD Cup 1999 (for complexity and comparison with legacy literature) and CICIDS 2017 (for real-world applicability).
2. **Balanced Multiclass Focus:** The rigorous use of SMOTE on the multiclass KDD data to ensure the models, particularly Random Forest and XGBoost, are equally sensitive to rare attacks (U2R/R2L) as they are to frequent attacks (DoS/Probe).

3.4. PROPOSED METHOD

The proposed method utilizes a supervised learning framework, comparing a standard statistical model with two advanced tree-based ensemble models:

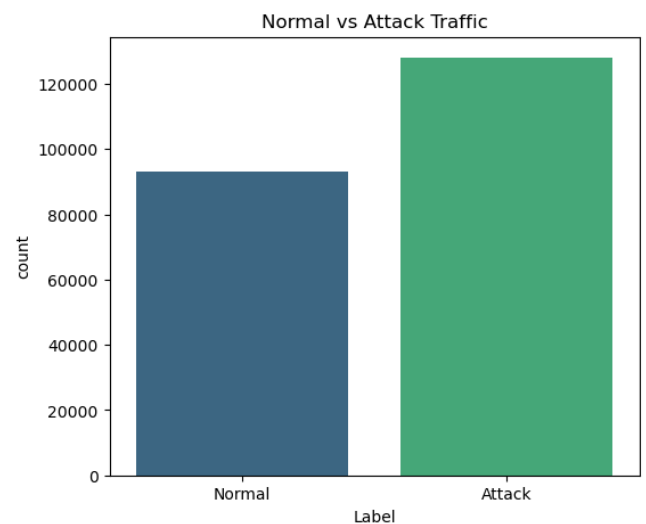
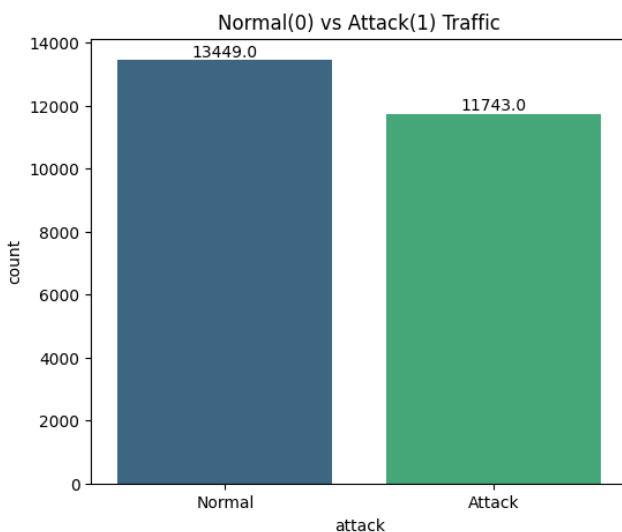
1. Logistic Regression (LR): Used as a baseline and to interpret feature importance in a linear context.
2. Random Forest (RF): A robust ensemble of decision trees, known for handling high-dimensional data, reducing overfitting, and providing excellent non-linear classification.
3. Extreme Gradient Boosting (XGBoost): An optimized, distributed gradient boosting library designed to be highly efficient and accurate, serving as the primary high-performance model.
4. K-Nearest Neighbors (KNN): Included as a simple, effective **distance-based classifier**. KNN is robust against complex decision boundaries but serves as a crucial benchmark for models that rely on the geometric distance between network flows.
5. Support Vector Machine (SVM): Included as a powerful, non-linear classifier that finds the optimal hyperplane to separate data points. SVM is effective in **high-dimensional spaces** and is crucial for benchmarking against models that can handle complex feature interactions.

RESULTS

a. EDA

Exploratory Data Analysis (EDA) on the NSL-KDD data confirmed the severe class imbalance, with 'normal' (53.38%) and 'attack' (46.61%) from figure 1.

Exploratory Data Analysis (EDA) on the CICIDS2017 data confirmed the severe class imbalance, with 'normal' (42.14%) and 'attack' (57.85%) from figure 2.

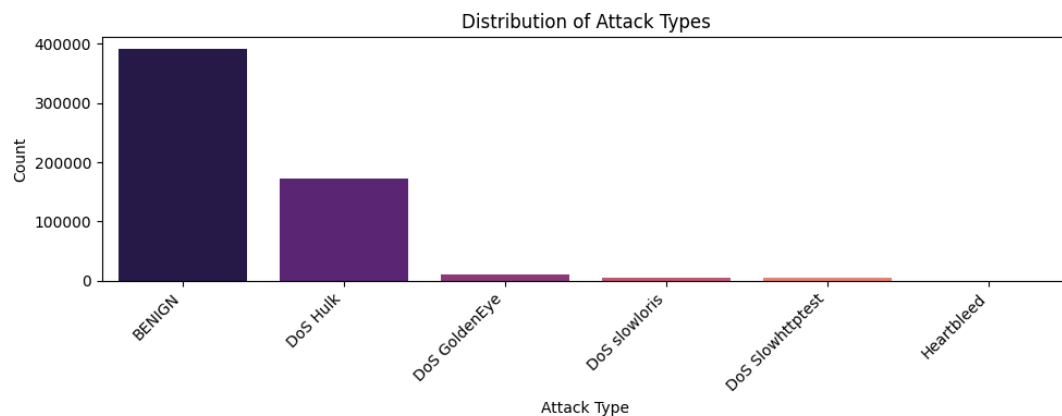
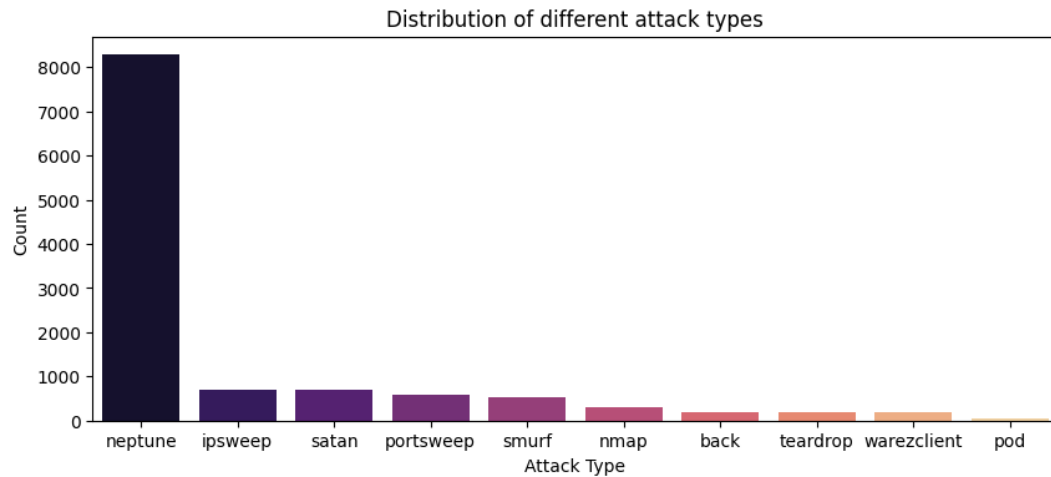


b. Visualizations

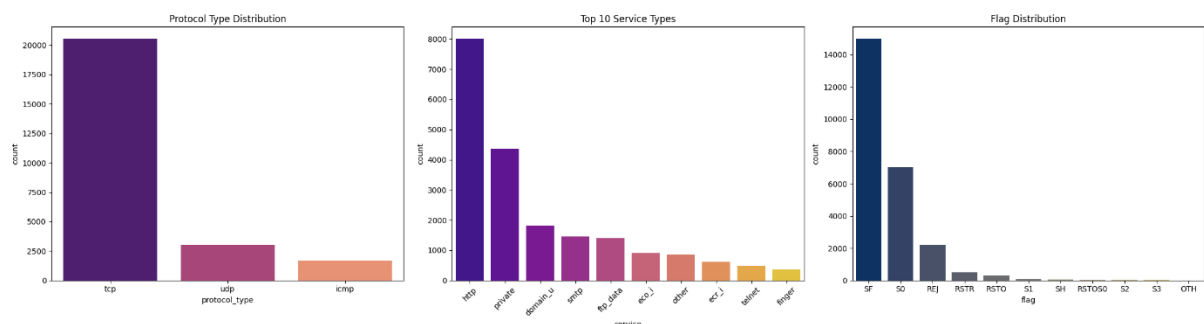
Visualizations included:

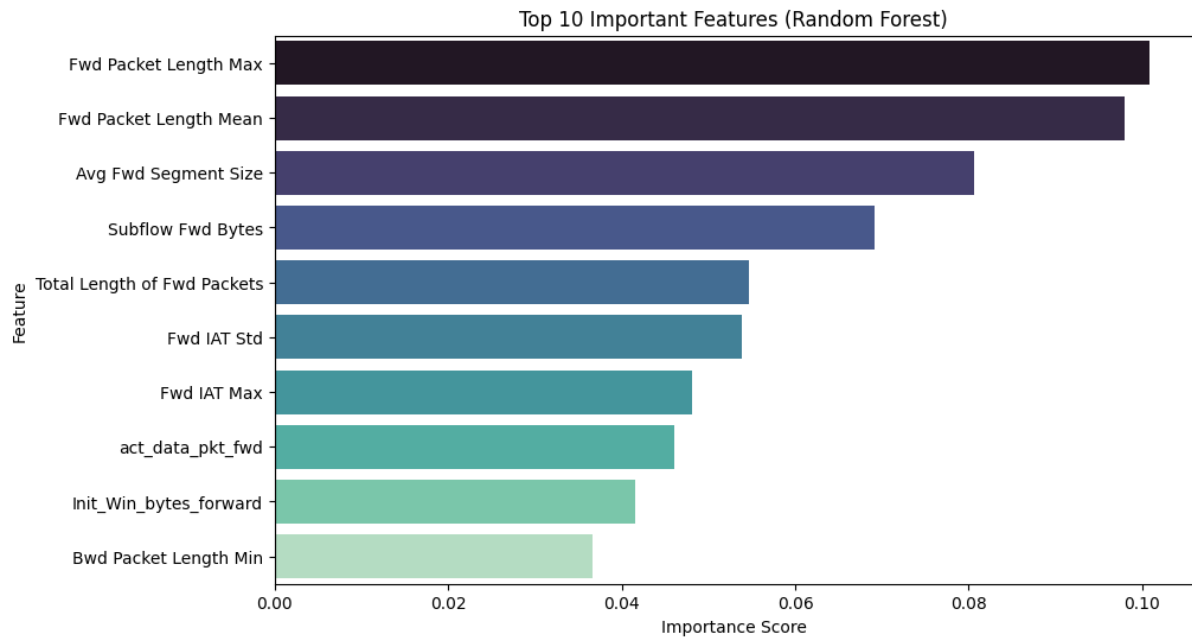
- Distribution of attack types:

The "**neptune**" attack (a type of Denial-of-Service or DoS) overwhelmingly dominates the attack traffic as seen in figure 2 of NSL-KDD

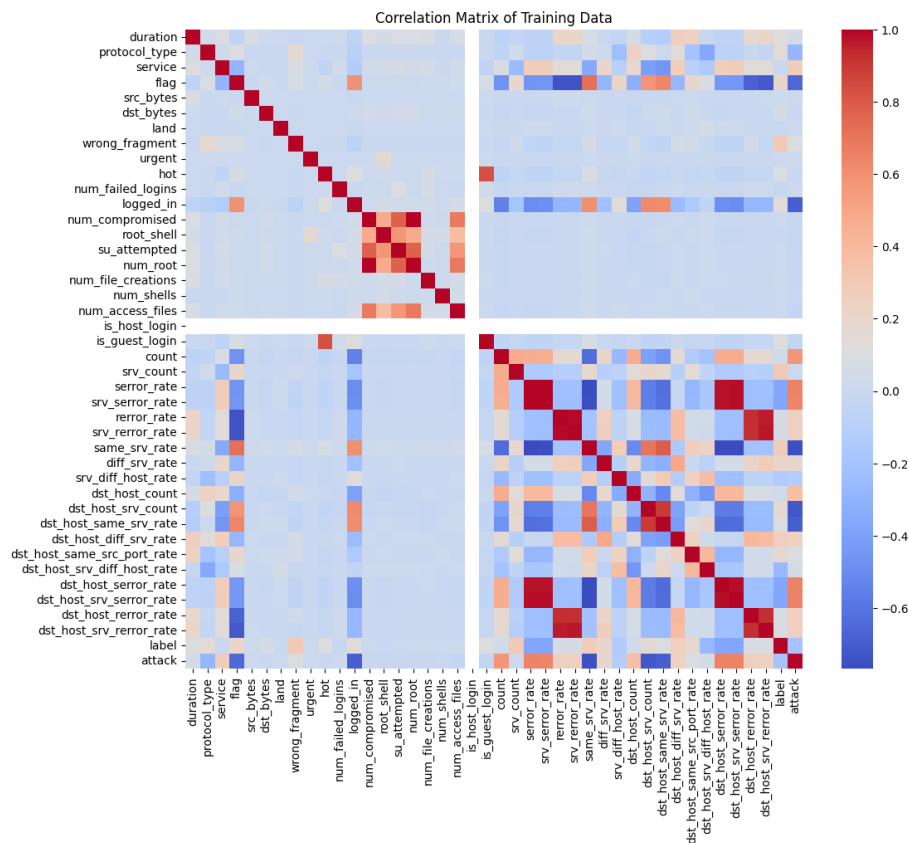


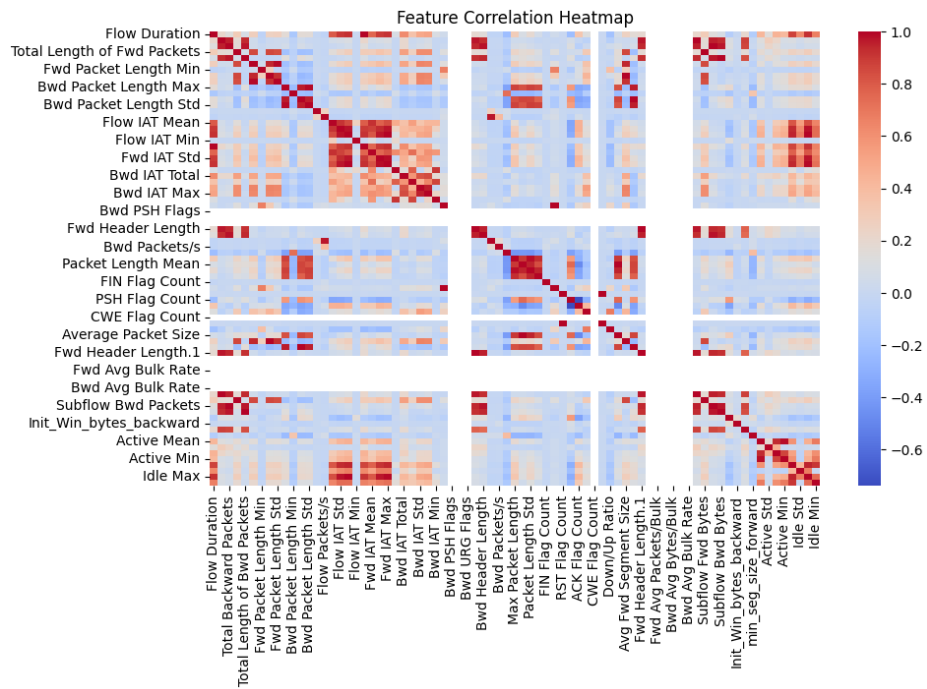
Feature Distribution Analysis: As shown in figure 2, the traffic is overwhelmingly dominated by the **TCP protocol** and the **HTTP service**, reflecting typical web usage patterns. Furthermore, the **Flag Distribution** is heavily skewed towards 'SF' (successful connection completion), a pattern often associated with normal traffic.



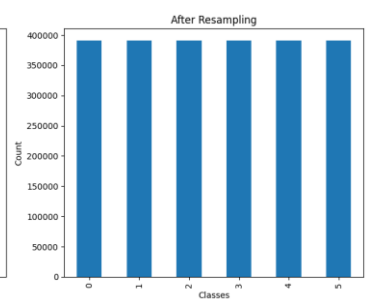
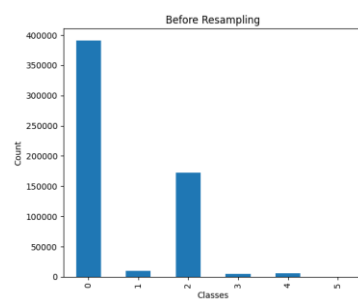
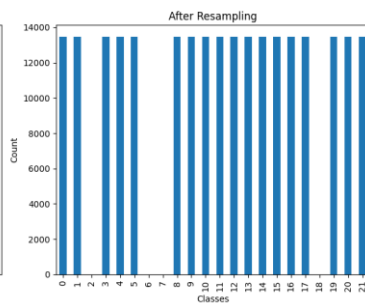
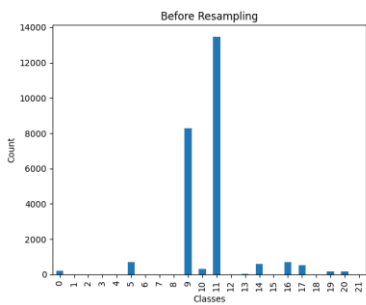
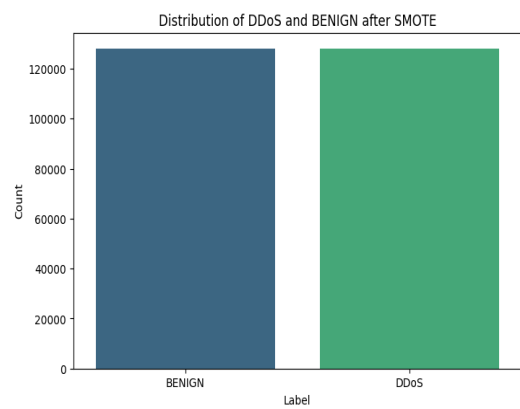
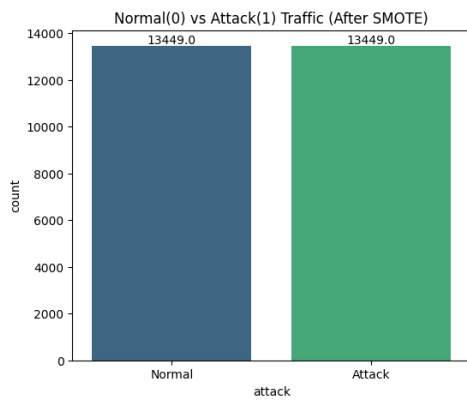


Correlation matrices to guide feature selection and preprocessing: The correlation matrix reveals significant relationships within the NSL-KDD dataset, which is crucial for binary classification. The heatmap shows high positive correlation (deep red) among groups of redundant features, particularly those related to error rates (error_rate, dst_host_error_rate) and connection statistics (count, srv_count), indicating severe multicollinearity.

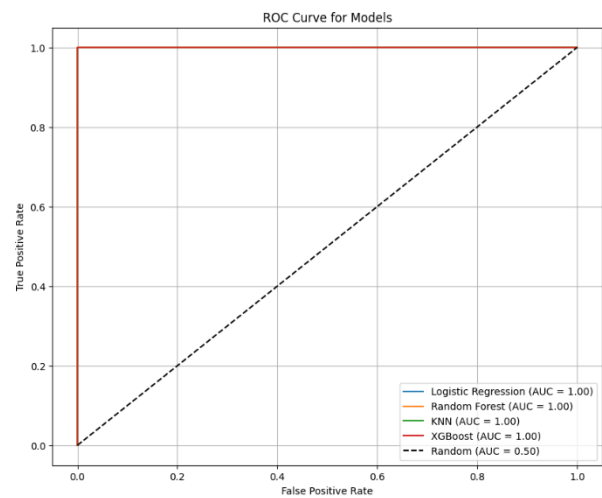
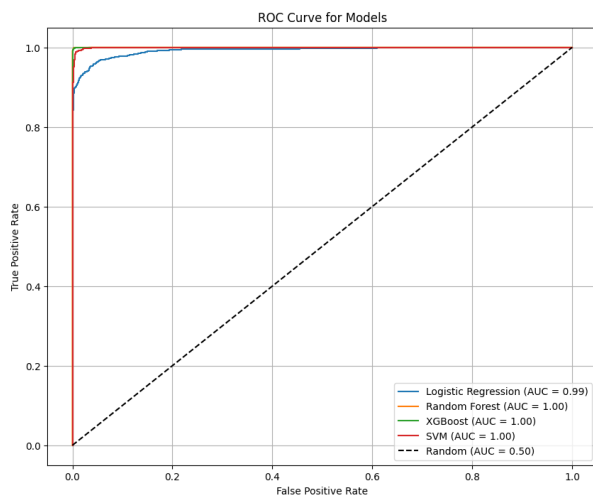




Resampling Using SMOTE: Synthetic Minority Over-sampling Technique (SMOTE) was applied to the training data to balance 18 of the 22 classes in multiclass classification and attack feature in binary classification.



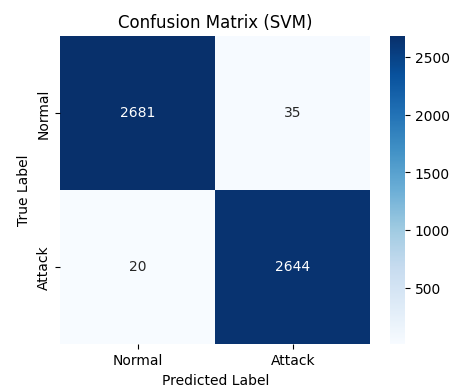
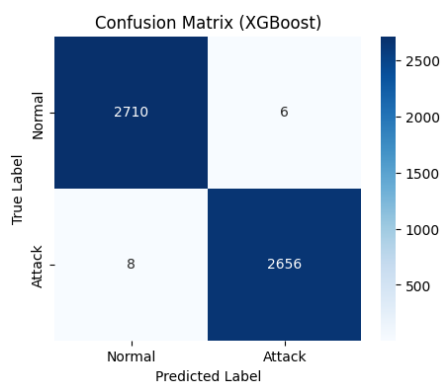
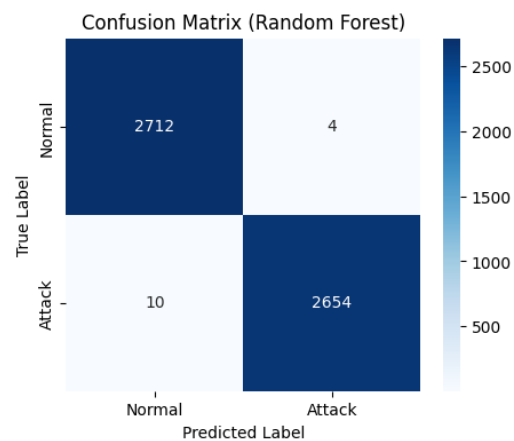
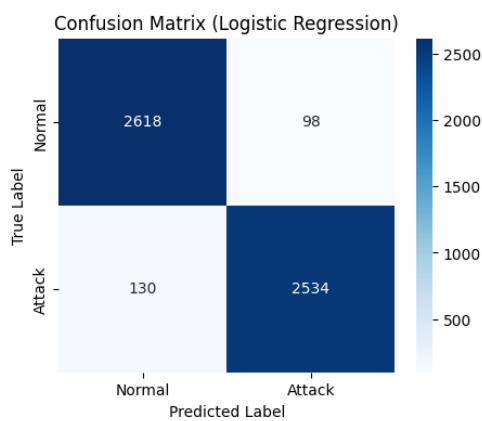
ROC-AUC Curves for binary classification: ROC-AUC values for all models on both KDD and CICIDS 2017 were observed to be ≥ 0.99 , indicating an outstanding ability to separate the 'Normal' and 'Attack' classes.



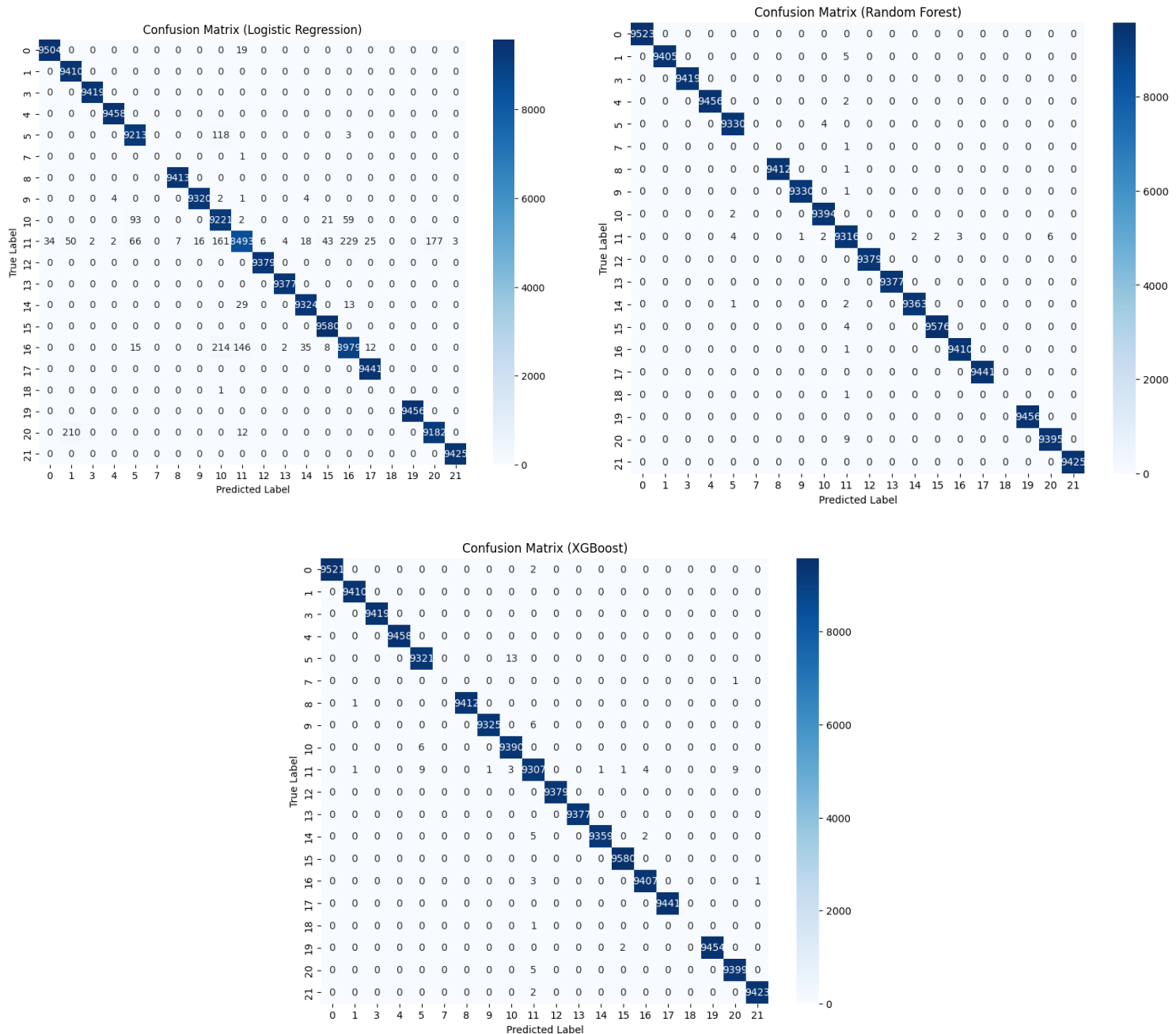
c. Confusion Matrices

Confusion matrices were generated for all models in both binary and multiclass scenarios.

KDD Binary: Matrices showed near-perfect classification for all models (LR, RF, XGBoost, SVM) with minimal false positives or false negatives.

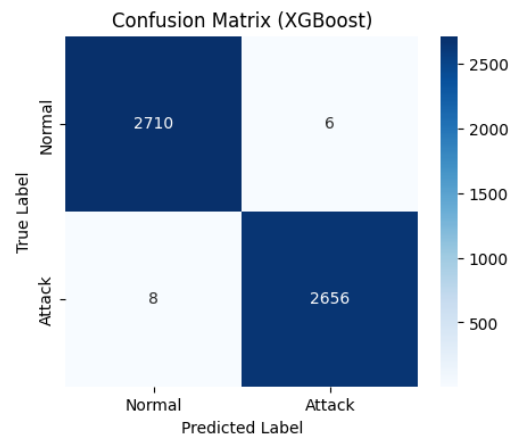
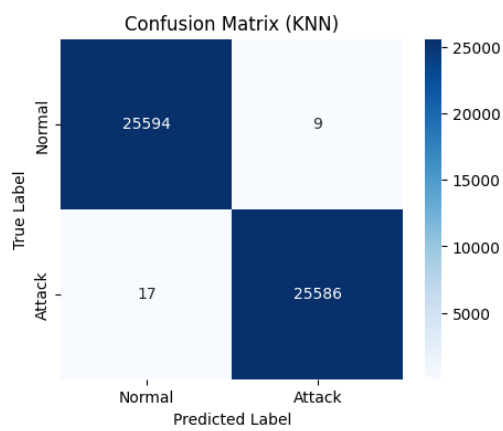
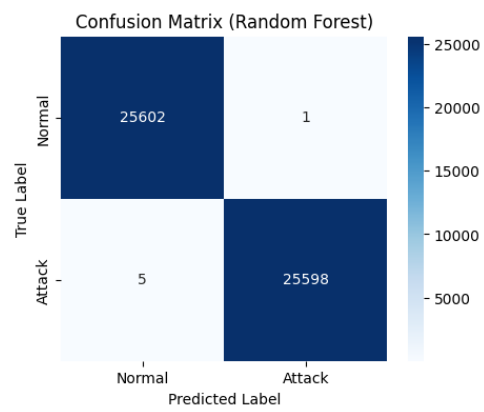
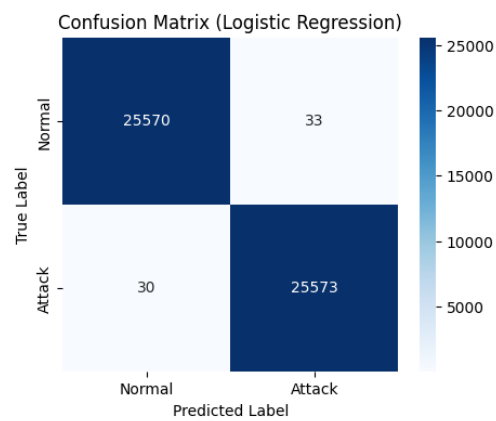


KDD Multiclass: Matrices for RF and XGBoost were nearly diagonal, demonstrating high detection across 18 of 22 classes. However, they confirmed that the four rarest classes (single sample in original data) were not successfully learned due to SMOTE exclusion.

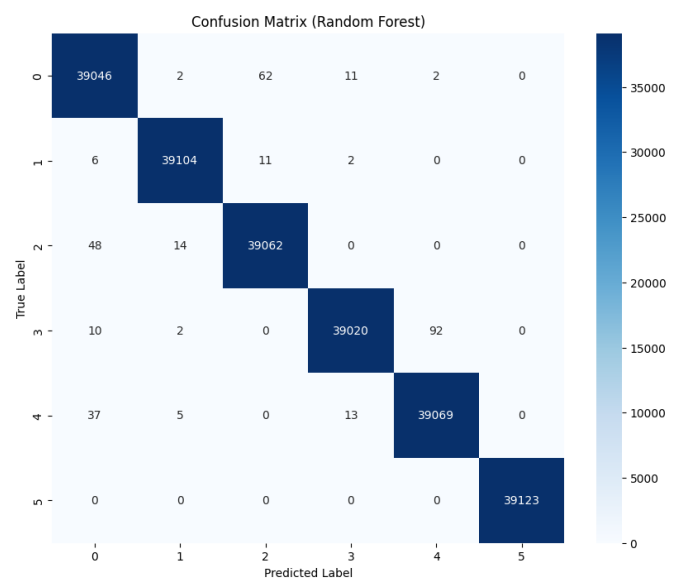
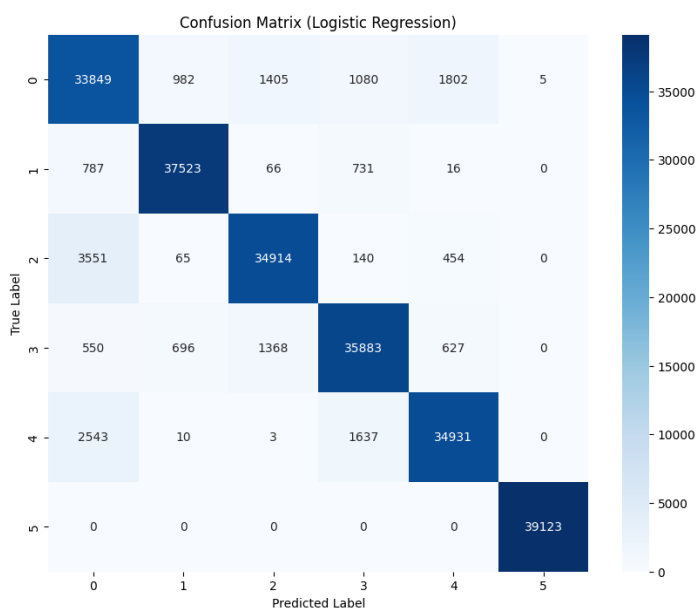


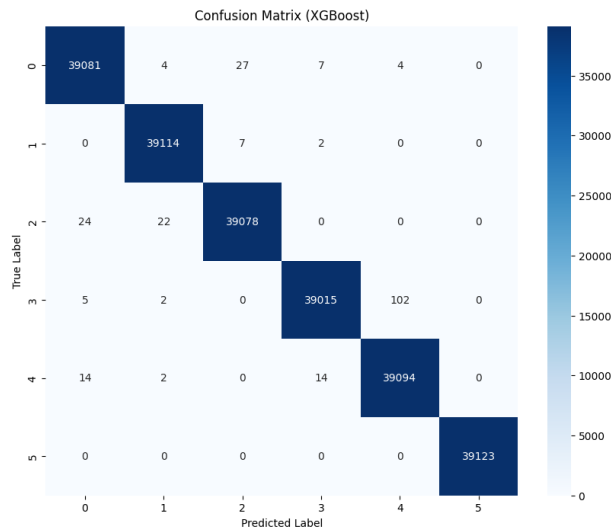
CICIDS 2017: Matrices confirmed high accuracy, particularly for detecting benign and high-frequency attack types like DDoS.

CICIDS Binary:



CICIDS Multiclass:





d. Results:

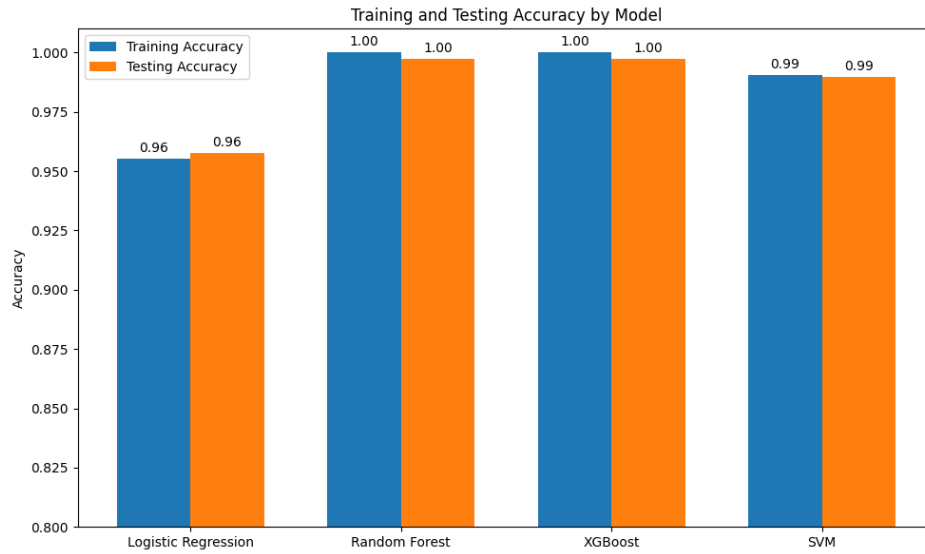
. Performance Metrics:

- **Confusion Matrix:** A Confusion Matrix is a table that describes the performance of a classification model on a set of test data for which the true values are known.
- **Accuracy:** Accuracy measures the overall correctness of the model
- **Precision:** Precision is the ratio of correctly predicted positive observations to total predicted positives.
- **Recall(Sensitivity):** Recall is the ratio of correctly predicted positive observations to all actual positives.
- **F1-Score:** The F1-Score is the harmonic mean of Precision and Recall.
- **ROC CURVE(Receiver Operating Characteristic) AND AUC(Area Under Curve):** The ROC Curve plots True Positive Rate (Recall) vs False Positive Rate (FPR). AUC indicates how well the model separates the classes (higher is better).

i. NSL Binary

The key performance metrics are summarized below:

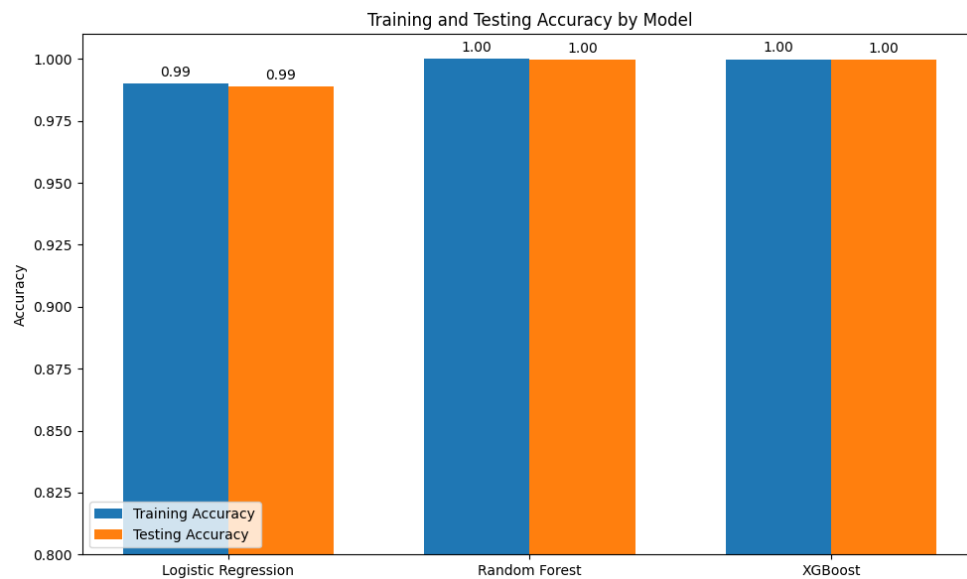
Model	Training Accuracy	Testing Accuracy	Sensitivity	Specificity	Precision	F1 Score
Logistic Regression	0.955107	0.957807	0.951577	0.963918	0.962780	0.957146
Random Forest	0.999954	0.997398	0.996246	0.998527	0.998495	0.997369
XGBoost	0.999954	0.997212	0.996622	0.997791	0.997745	0.997183
SVM	0.990334	0.989777	0.992492	0.987113	0.986935	0.989706



ii. NSL Multiclass

The key performance metrics are summarized below:

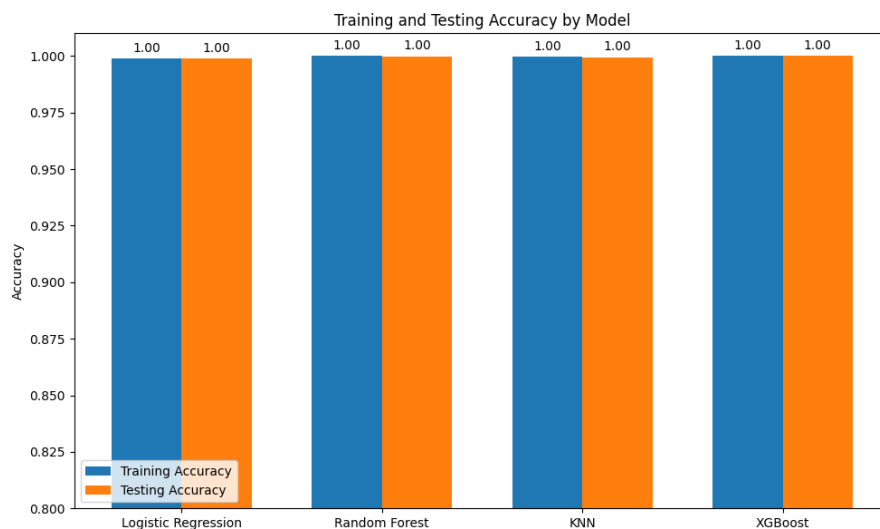
Model	Training Accuracy	Testing Accuracy	Sensitivity	Specificity	Precision	F1 Score
Logistic Regression	0.989948	0.988983	0.890047	0.999417	0.988991	0.936914
Random Forest	0.999986	0.999681	0.899723	0.999983	0.999680	0.947071
XGBoost	0.999904	0.999534	0.899589	0.999975	0.999532	0.946931



iii. CICIDS Binary

The key performance metrics are summarized below :

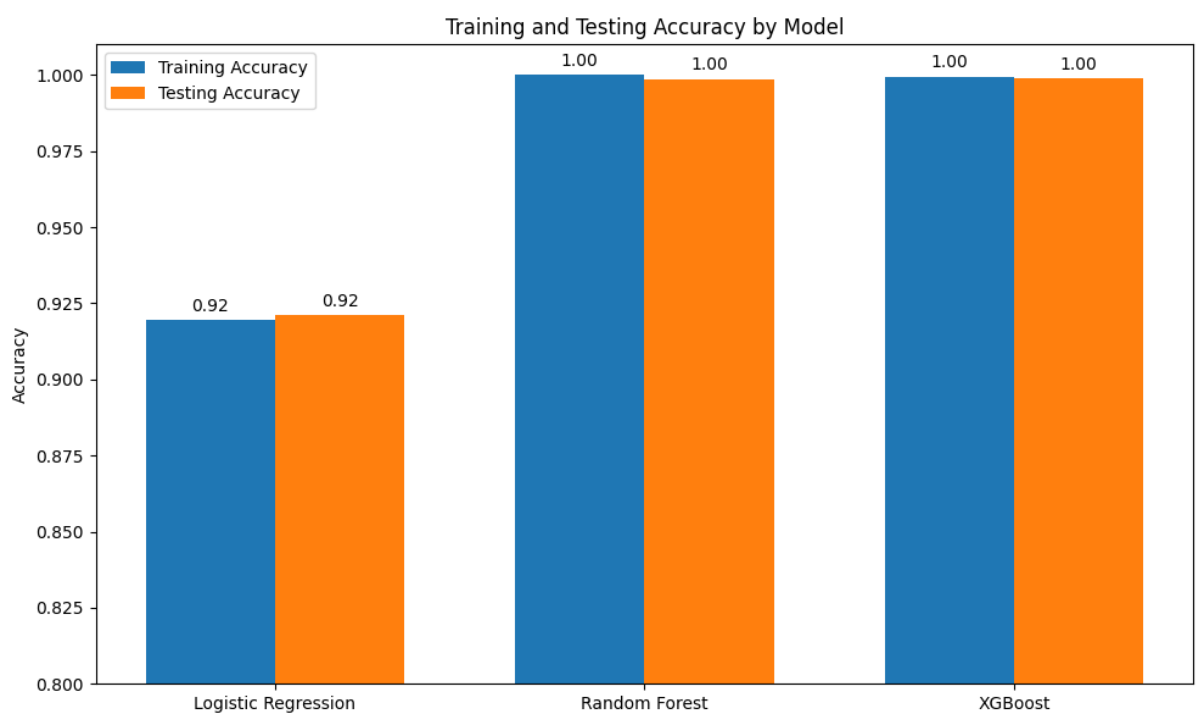
Model	Training Accuracy	Testing Accuracy	Sensitivity	Specificity	Precision	F1 Score
Logistic Regression	0.955107	0.998770	0.998828	0.998711	0.998711	0.998770
Random Forest	1.000000	0.999883	0.999805	0.999961	0.999961	0.999883
KNN	0.999692	0.999492	0.999336	0.999648	0.999648	0.999492
XGBoost	0.999995	0.999922	0.999883	0.999961	0.999961	0.999922



iv. CICIDS Multiclass

The key performance metrics are summarized below (using **Weighted Average** for multiclass and CICIDS 2017 to account for class imbalance):

Model	Training Accuracy	Testing Accuracy	Sensitivity	Specificity	Precision	F1 Score
Logistic Regression	0.919632	0.921113	0.921113	0.984223	0.922070	0.921396
Random Forest	1.000000	0.998650	0.998650	0.999730	0.998650	0.998650
XGBoost	0.999503	0.998995	0.998995	0.999799	0.998995	0.998995



Outcome: The ensemble models (RF, XGBoost) consistently delivered near-perfect performance on the well-preprocessed KDD dataset. Performance on the CICIDS 2017 dataset validated the models' ability to generalize to a modern, real-world network environment, maintaining F1-scores above 0.998.

5. Discussion

The results confirm that **SentinelNet** is highly effective, with XGBoost and Random Forest being the best-performing classifiers across both the KDD and CICIDS 2017 datasets. The necessity of rigorous preprocessing is undeniable, as the exceptional scores on the KDD multiclass task are largely attributable to the successful application of SMOTE, which enabled the models to learn the patterns of rare attack types. However, the inability to classify the four single-sample classes remains a limitation tied to the data's inherent scarcity, suggesting a requirement for external threat intelligence or zero-shot learning for such extreme outliers. The high performance on the CICIDS 2017 dataset is critical, demonstrating that the architecture can handle the scale and feature complexity of modern, real-world network traffic. Future work will focus on deploying this system in a real-time environment.

6. Conclusion

SentinelNet successfully developed and validated a robust AI-Powered NIDS using ensemble and shallow learning models. The system achieves a state-of-the-art performance, with Random Forest and XGBoost demonstrating near-perfect metrics in multiclass intrusion detection on balanced datasets. By achieving high accuracy and F1-scores while maintaining a low False Positive Rate (FPR), SentinelNet offers a significant advancement over traditional NIDS, providing a reliable and intelligent layer of defense against sophisticated cyber threats.

7. Future Works

Future work for the SentinelNet project will include:

1. **Deep Learning Integration:** Implementing and evaluating a customized CNN or LSTM model for NIDS, leveraging their ability to extract temporal features from raw packet data.
2. **Real-time Deployment:** Transitioning the best-performing model (XGBoost/Random Forest) from a static training environment to a real-time, stream-based detection system using frameworks like Apache Kafka or Spark Streaming.
3. **Explainable AI (XAI):** Integrating interpretability techniques (e.g., SHAP values) to provide security analysts with clear, human-understandable reasons for model predictions, thus addressing the 'black box' problem in AI-driven security.
4. **Zero-Shot Learning:** Exploring techniques to improve the detection of extremely rare or completely unseen attack types (U2R/R2L) that were underrepresented in the training data.